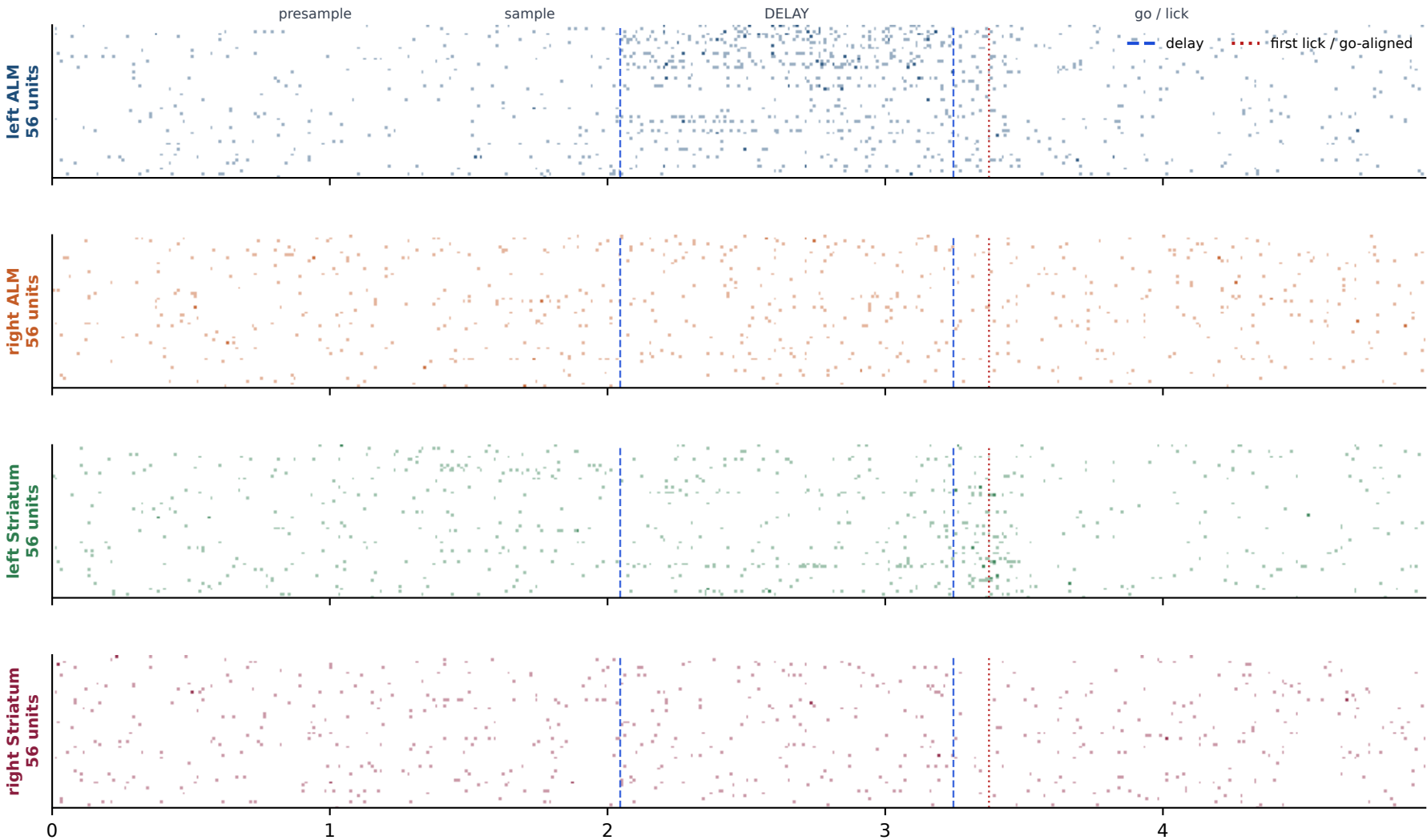


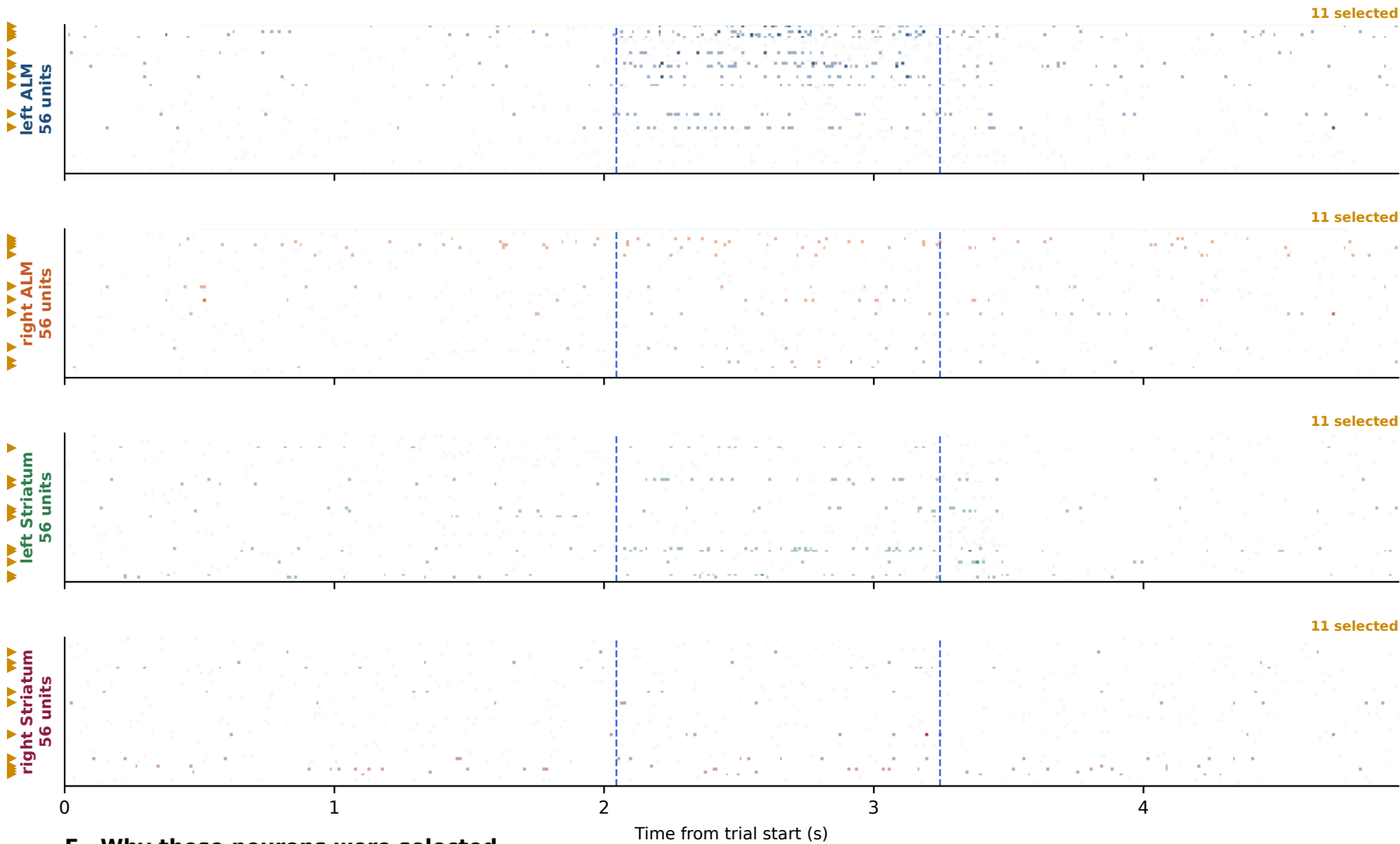
Figure 1 Causal predictive neuron selection across ALM and striatum

Single right-lick trial · four-region raster · delay context used to forecast lick-period activity · selected units marked in gold

A All recorded units



B Units retained by SPEC selection (same trial; unselected faded)



F Why these neurons were selected

Composite score = 0.28·z(attention) + 0.22·z(prediction gain) + 0.22·z(d') + 0.16·z(delay→lick r) + 0.12·z(TF selectivity). Top 18% per region kept. Silent units (delay rate < 0.4 Hz) are penalized.

right Striatum · unit 219 · prefers Left · score 3.45

- Choice-discriminative delay rate (d'=3.96 vs other actions; prefers Left).
- SPEC-TCNN neuron attention is high (a=0.146); the unit is used as past context.
- Occluding this unit raises lick-raster prediction error (ΔMSE gain=0.070).

left Striatum · unit 156 · prefers Right · score 2.83

- Choice-discriminative delay rate (d'=3.61 vs other actions; prefers Right).
- SPEC-TCNN neuron attention is high (a=0.132); the unit is used as past context.
- Occluding this unit raises lick-raster prediction error (ΔMSE gain=0.091).

left Striatum · unit 157 · prefers Right · score 3.09

- Choice-discriminative delay rate (d'=4.22 vs other actions; prefers Right).
- SPEC-TCNN neuron attention is high (a=0.241); the unit is used as past context.
- Occluding this unit raises lick-raster prediction error (ΔMSE gain=0.037).

right ALM · unit 63 · prefers Left · score 2.65

- Choice-discriminative delay rate (d'=8.64 vs other actions; prefers Left).
- SPEC-TCNN neuron attention is high (a=0.673); the unit is used as past context.
- Ground-truth type: persistent delay-choice cell — the ALM preparatory motif of Li/Inagaki/Svoboda.

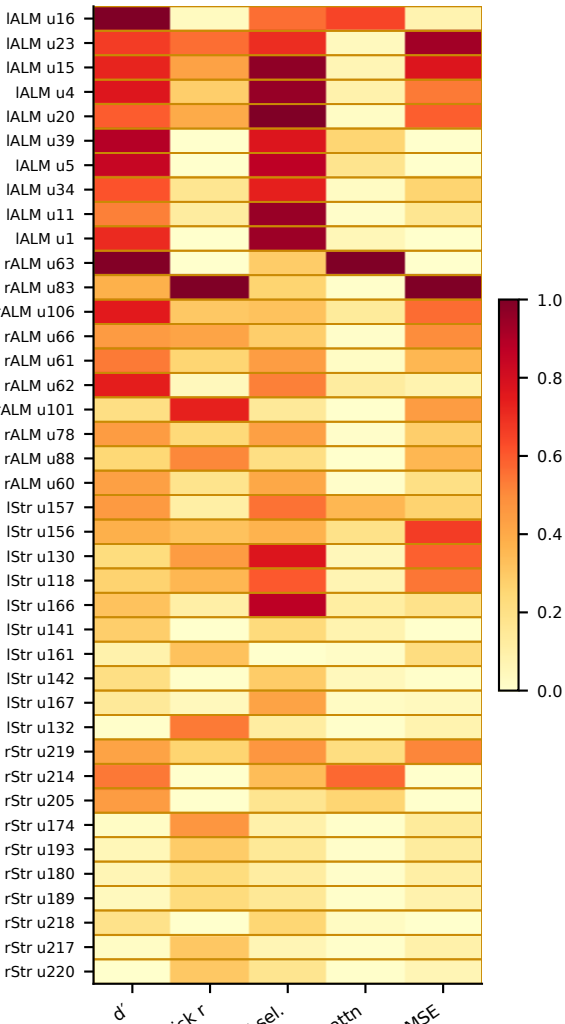
right Striatum · unit 214 · prefers Left · score 2.85

- Choice-discriminative delay rate (d'=4.93 vs other actions; prefers Left).
- SPEC-TCNN neuron attention is high (a=0.385); the unit is used as past context.
- Ground-truth type: persistent delay-choice cell — the ALM preparatory motif of Li/Inagaki/Svoboda.

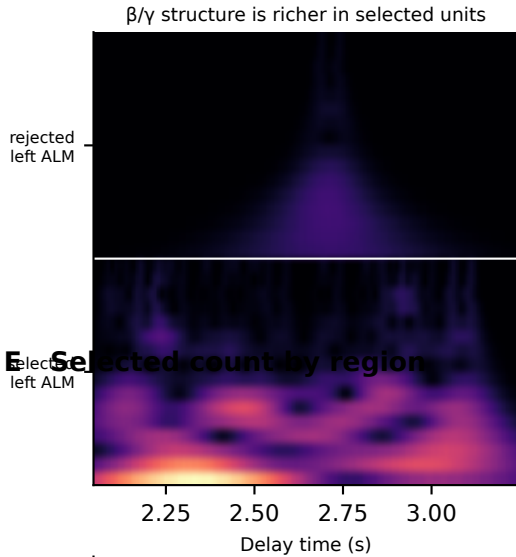
left ALM · unit 16 · prefers Right · score 2.43

- Choice-discriminative delay rate (d'=8.73 vs other actions; prefers Right).
- SPEC-TCNN neuron attention is high (a=0.436); the unit is used as past context.
- Occluding this unit raises lick-raster prediction error (ΔMSE gain=0.011).

C Selection scores



D Delay CWT, Selected vs rejected



E Selected count by region

